

GROUP ASSIGNMENT

SUBMITTED BY:

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SUBMITTED TO:

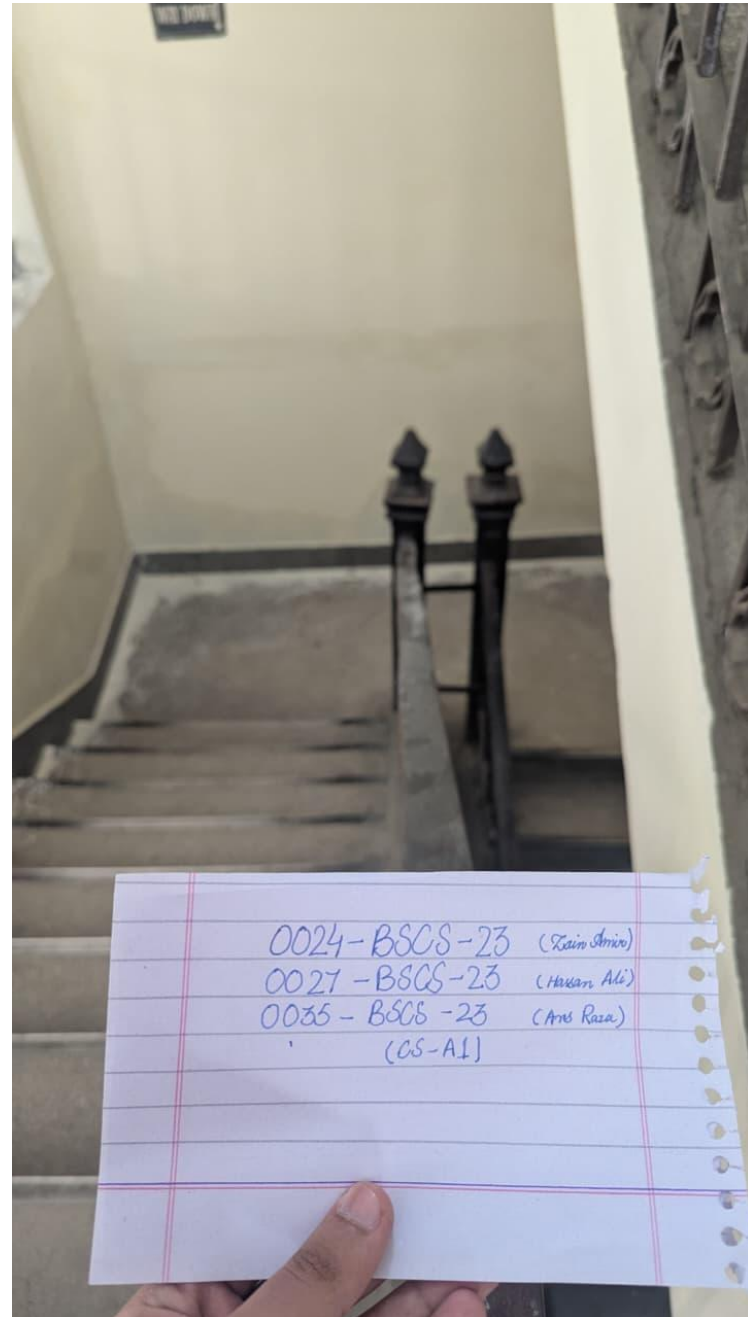
Sir Awais Qasim



1) Stair railing:

DESCRIPTION:

The stair railing provides physical support while climbing stairs. It reduces strain on the legs while going up or down. It maintains balance and posture and prevents falls and injuries, making movement easier for all users.



2) Hydraulic Door Closer:

DESCRIPTION:

This is the picture we took in main lab and this door closure relates to ergonomics because:

- It reduces the need for manual force to close heavy doors.
- It prevents sudden slamming that reduces injury risk.
- It improves usability for frequent entry/exit areas.

The hydraulic door closer automatically controls the speed and force of door closing.

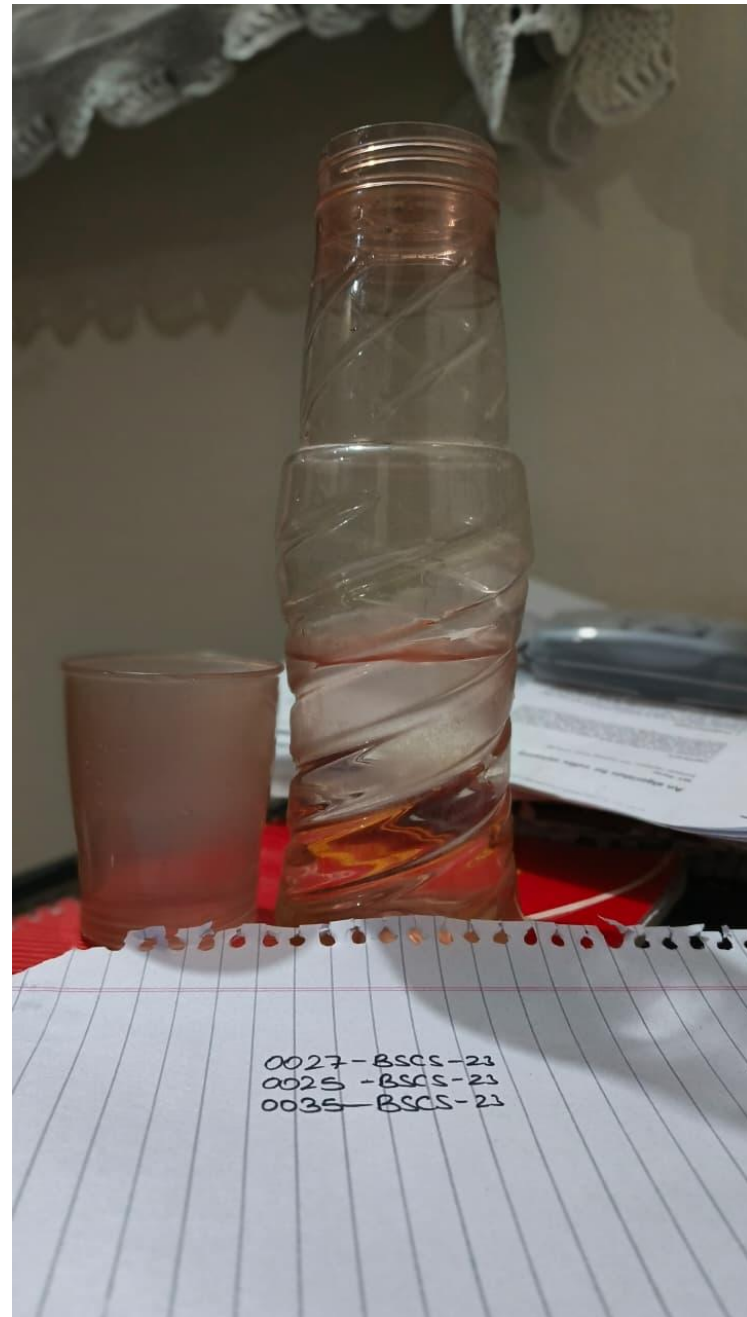


3) Ergonomic Water Bottle:

DESCRIPTION:

Unlike regular/normal water bottles, this design has:

- an easy grip shape.
- It reduces wrist strain while holding
- Its grip-friendly shape enhances comfort and promotes hydration, which supports physical well-being.



4) Ergonomics Chair with lumbar head support:

DESCRIPTION:

An ergonomics chair is designed with adjustable lumbar support to maintain the natural inward curve of the lower back. The headrest supports the cervical spine and reduces neck strain. Lack of head and lumbar support increases pressure on the spine.

Adjustable height and armrests allow proper alignment of knees at 90 degrees, promoting healthy posture and reducing musculoskeletal stress.

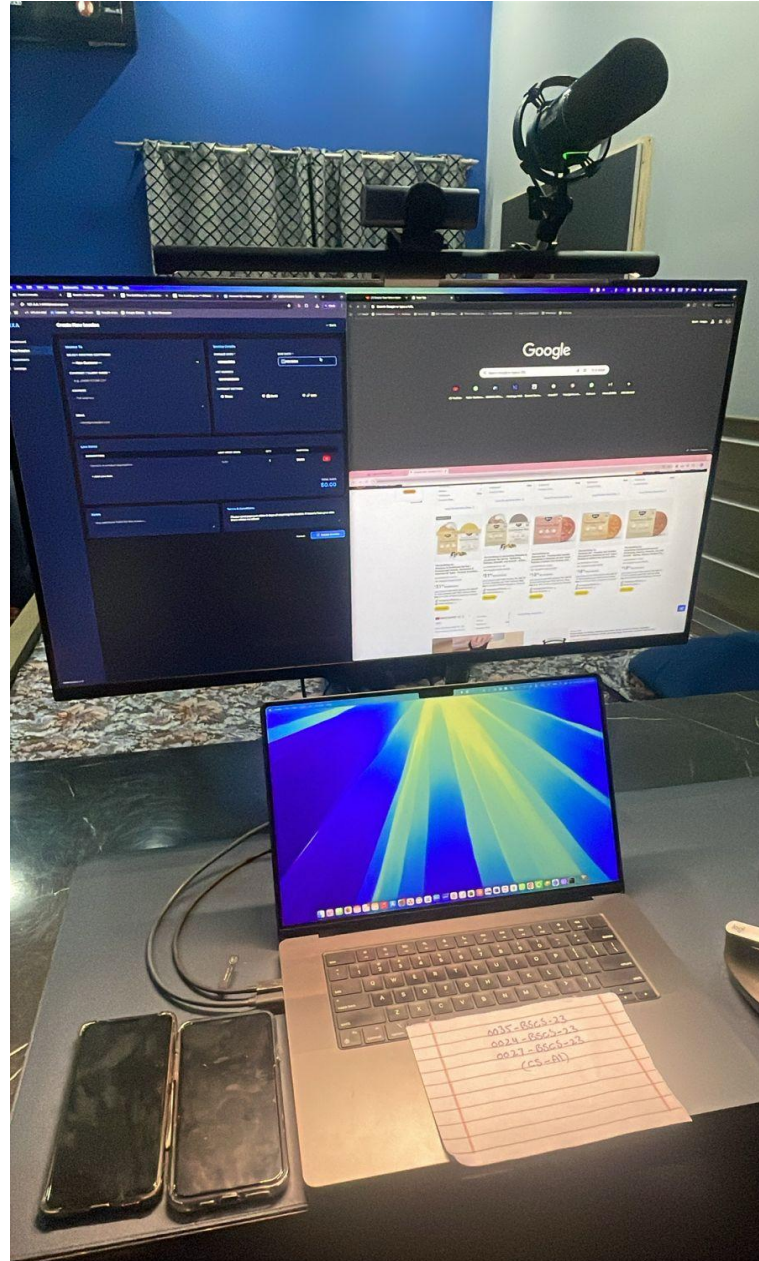


5) **Multi-Monitor Setup
(Dual/Triple Monitor Display):**

DESCRIPTION:

A multi-monitor setup allows user to distribute task across multiple screens, reducing constant window switching , excessive scrolling , and repeated neck.

This increases visual strain and reduces productivity. Proper screen alignment reduces excessive cervical rotation and visual strain during prolonged company use.



6) Ergonomic Laptop Stand:

DESCRIPTION:

An ergonomic laptop stand helps in reducing neck strain by placing the laptop screen to eye, allowing the user to maintain a neutral head and spine position.

This reduces cervical spine strain and improves overall posture during extended computer use.

It reduces wrist and shoulder tension by aligning keyboard and screen optimally.

It fits various laptop sizes, making it versatile for different workspaces.



7) 3-Pin to 2-Pin Laptop Charger Adapter:

DESCRIPTION:

- It converts 3-pin charger to fit 2-pin sockets, improving plug compatibility.
- It enhances user convenience and safety with adaptable charging.
- It reduces physical hassle and unsafe workarounds at workstations.
- It promotes ergonomic work environment by minimizing effort.
- It improves electrical safety for laptop users.
- It Suitable for various laptop models and outlets.
- It simplifies workspace setup and charging flexibility.



8) Sewing Machine Foot Pedal Accelerator:

DESCRIPTION:

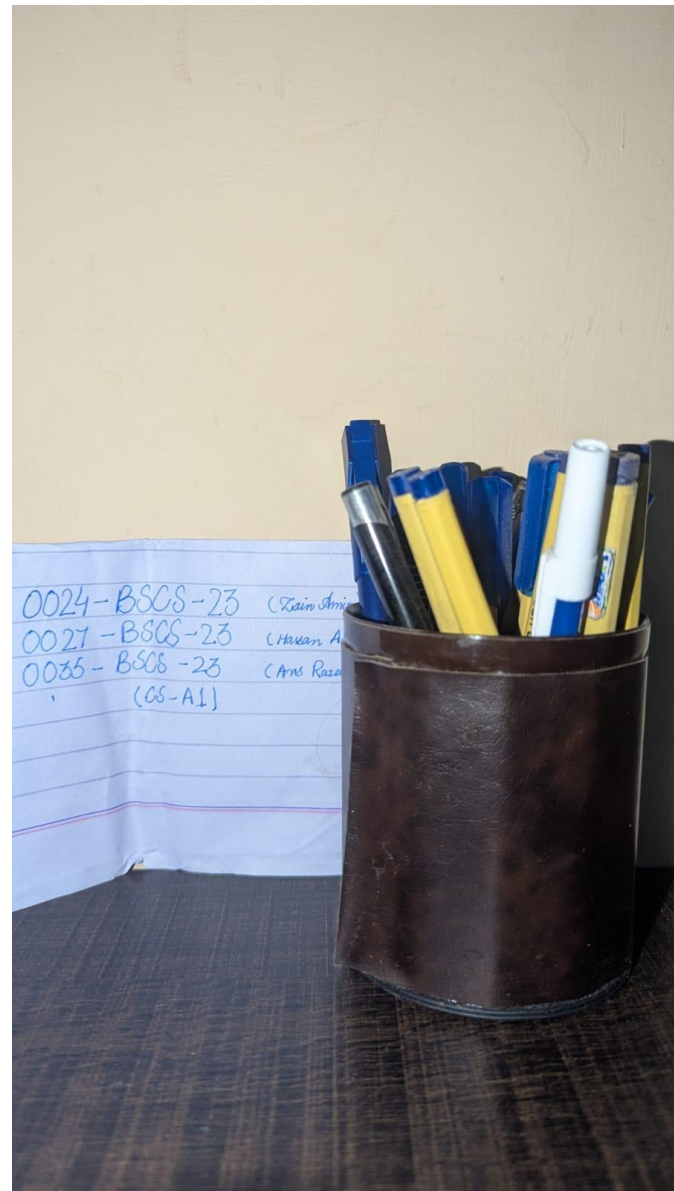
- It enhances user comfort and efficiency with convenient foot control.
- It reduces physical strain and fatigue during operation.
- It allows smooth, effortless machine activation.
- It Improves workstation ergonomics and safety.
- It promotes adaptable work practices and reduces risk of injury.
- It supports prolonged use with comfort-focused design.
- It simplifies sewing machine operation for better workflow.



9) Leather Pen Holder:

DESCRIPTION:

- An ergonomic desk accessory that organizes pens and markers for easy access, reducing clutter.
- Promotes comfortable workstation setup by keeping writing tools within reach, minimizing hand movement.
- Encourages efficient work practices by providing a stable, tidy storage solution.
- Enhances user comfort and productivity through simple, accessible tool arrangement.
- Supports ergonomic posture by preventing unnecessary bending or searching for items.



10) Multi-Media projector:

DESCRIPTION:

- Adjustable Projection Angle:
Can be tilted and rotated to reduce glare and improve image positioning
- Keystone Correction:
Automatically adjusts image shape to prevent distortion and reduce eye strain
- Low Noise Level:
Designed to minimize background noise and distractions
- Easy-to-Use Interface:
Intuitive controls and wireless connectivity make it simple to switch between devices
- Flexible Placement Options:
Can be ceiling-mounted, placed on a table, or used with a mobile stand

